

Leeuwenhoek, Anton van (1632–1723)

Dutch microscopist often considered the father of microbiology. Originally employed in the textile trade, Leeuwenhoek built and used microscopes, becoming the first to observe single-celled organisms, including bacteria and protozoa, as well as muscle fibres and blood flow in capillaries.

Leibniz, Gottfried von (1646–1716) German philosopher and mathematician who made major contributions to physics, metaphysics, optics, logic, statistics, mechanics, and technology. Leibniz developed calculus independently of Isaac Newton, built a calculating machine, and refined the binary system, which forms the foundation of digital technology. He published no major philosophical treatises.

Lenard, Philipp (1862–1947) German physicist awarded the 1905 Nobel Prize in Physics for his research on cathode rays. A professor at four German universities, Lenard supported Nazi doctrine and denounced “Jewish” science, including Einstein’s theory of relativity.

Liebig, Justus von (1803–73) German chemist whose pioneering work in the fields of organic chemistry, biochemistry, and agriculture contributed to the establishment of the fertilizer industry. Appointed professor at Giessen University at 21 years old, Liebig was the first to establish the laboratory-based teaching methodology that spread to the US and the rest of Europe.

Lind, James (1716–94) Scottish physician who tried to eradicate scurvy from the British navy by introducing citrus juice to the shipboard diet. Although the Navy was slow to adopt his ideas, he also introduced fumigation below decks, better hygiene for sailors, and the distillation of seawater into drinking water.

Linnaeus, Carolus (Carl von Linné), Swedish naturalist (1707–78) See p.139

Lippershey, Hans (c.1570–c.1619) Dutch eyeglass-maker commonly credited with inventing the telescope. In 1608, Lippershey sold his invention to the Dutch government for use in warfare. Later astronomers, notably Galileo, recognized the telescope’s importance for science. A planet and lunar crater were named after Lippershey.

Lister, Joseph (1827–1912) British surgeon and founder of antiseptic medicine. A professor and president of the Royal Society, Lister pioneered the principle of bacteria prevention during surgery, using carbolic acid to sterilize surgical instruments, and keeping postoperative wounds clean.

Lockyer, Joseph (1836–1920) English astronomer known for discovering the element helium in the Sun’s atmosphere and naming it. Originally a civil servant, Lockyer observed solar prominences in the Sun’s chromosphere, devised the spectroscopic observation of sunspots, and founded the periodical *Nature*.

Lodge, Oliver (1851–1940) English physicist noted for his pioneering work in wireless telegraphy. Lodge is best known for improving detector devices for transcribing Morse code radio waves onto paper. A keen promoter of spiritualism, Lodge also received patents for several wireless inventions.

Lomonosov, Mikhail, Russian chemist, physicist, geographer, and astronomer (1711–65) See p.145

Lonsdale, Kathleen (1903–71) Irish crystallographer who developed X-ray techniques to investigate chemical structures. Lonsdale established the hexagonal shape of carbon atoms in benzene, and determined the structure of hexachlorobenzene. In 1945, Lonsdale became the first woman to be elected a Royal Society fellow.

Lord Kelvin [see Thomson, William]

Lorentz, Hendrik Antoon (1853–1928) Dutch physicist who shared the 1902 Nobel Prize in Physics with Pieter Zeeman for their work on electromagnetic radiation. The first to describe the force of charged particles within an electromagnetic field, Lorentz’s analyzed how events may be perceived at different times in different frames of reference, and developed the transformation equations that underpinned Einstein’s theory of relativity.

Lorenz, Konrad (1903–89) Austrian founder of ethology and co-winner of the 1973 Nobel Prize in Physiology or Medicine for research on animal behavior. Lorenz is noted for studying imprinting in birds, and also animal aggression, which he argued is motivated purely by survival.

Lovelace, Ada, British mathematician (1815–52) See p.197

Lovelock, James (1919–) English chemist best known for his 1979 Gaia hypothesis, which proposes that Earth is a living organism “maintained and regulated by life on the surface.” An ardent environmentalist, Lovelock invented the electron capture detector to reveal chlorofluorocarbons in the atmosphere.

Lyell, Charles (1797–1875) Scottish geologist who proposed that the geological features of Earth’s surface were shaped by processes still operating at the same rate as in the past. His theory of uniformitarianism, presented in *Principles of Geology* (1830–33), was vital for Charles Darwin’s theories because it provided a greatly expanded time frame for Earth’s history.

Malpighi, Marcello (1628–94) Italian physician and biologist who founded the science of microscopic anatomy through his study of plant and animal tissue. Personal physician to Pope Innocent XII and a pioneer of brain anatomy, Malpighi named capillaries, contributed to embryology, discovered taste buds, and investigated the anatomy of frog lungs.

Malthus, Thomas Robert (1766–1834) English economist, clergyman, and philosopher who argued that natural growth in human population will always outstrip the food supply. To preserve humanity, Malthus proposed strict limits on reproduction or that overpopulation be left to be checked by war or famine. His theory, known as Malthusianism, greatly influenced social, political, and economic thought.

Mandelbrot, Benoit (1924–2010) Polish-born French-American mathematician who introduced the Mandelbrot set and fractal geometry, which shows how visual complexity can be created from simple shapes. A professor at Yale University, he examined many phenomena which, like a rocky

coastline, seem equally rough or jagged however close or far away you get.

Marconi, Guglielmo (1874–1937) Italian physicist and inventor of the wireless telegraph. Marconi sent the first wireless signal across the English channel in 1896 and across the Atlantic Ocean in 1902. He shared the 1909 Nobel Prize in Physics with Ferdinand Braun and helped develop shortwave wireless communication.

Margulis, Lynn, American biologist (1938–2011) See p.300

Maudslay, Henry (1771–1831) English inventor and engineer considered a founding father of the machine-tool industry. Originally a locksmith’s apprentice, Maudslay invented many important machines during the Industrial Revolution, such as the metal lathe, marine engines, and methods for desalinating seawater and printing calico cloth.

Maxwell, James Clerk, British physicist (1831–79) See p.209

Mayer, Julius Robert von (1814–78) German physicist, physician, and early founder of thermodynamics. Mayer was the first to determine the mechanical equivalent of heat, although this was credited to James Joule. He also described oxidation as the primary energy source for living creatures.

Mendel, Gregor (1822–84) Austrian monk and botanist whose plant experiments laid the foundation for modern genetics. Experimenting with garden peas, Mendel discovered that the characteristics of an individual are controlled by hereditary factors, now called genes. The significance of Mendel’s findings was not recognized until the early 20th century.

Mendeleev, Dmitri, Russian chemist (1834–1907) See p.211

Mercator, Gerardus, Flemish cartographer (1512–94) See p.73

Michell, John (1724–93) English clergyman, astronomer, and pioneer of seismology. In 1760, Michell proposed that earthquakes were wave motions in Earth’s crust, and in 1790, he created a torsion balance to measure Earth’s density.

Michelson, Albert Abraham (1852–1931) Polish-American physicist who accurately measured the speed of light. His experiments with Edward Morley to detect the drift of an ether were important for the understanding of Einstein’s relativity theory. He was awarded the Nobel Prize in Physics in 1907, the first American to receive a scientific Nobel prize.

Millikan, Robert (1868–1953) American physicist awarded the 1923 Nobel Prize in Physics for measuring the electrical charge of the electron with his oil-drop experiments. Millikan also confirmed Einstein’s photoelectric equation and conducted studies on the nature of cosmic rays, X-rays, and electric constants.

Mitchell, Maria (1818–89) First American woman to work as a professional astronomer, and discoverer of a comet named after her. Mitchell became director of the Vassar Female College’s Observatory in 1865, and she also founded the Association for the Advancement of Women.

Montagu, Lady Mary Wortley, English writer (1689–1762) See p.131

Morgan, Thomas Hunt (1866–1945) American geneticist and biologist whose research on the *Drosophila* fruit fly helped establish genetics. Morgan’s experiments showed that genes are arranged on chromosomes and are responsible for hereditary traits. He received the Nobel Prize in Physiology or Medicine in 1933.

Moseley, Henry (1887–1915) English physicist who used X-ray spectroscopy to prove the theory of atomic numbers. Working under Ernest Rutherford at Manchester University, Moseley confirmed physically the atomic numbers for elements that had been derived chemically. Like Mendeleev, his research enabled him to predict elements for gaps in the periodic table. Moseley was killed in World War I.

Murchison, Roderick Impey (1792–1871) Scottish geologist best known for establishing the Silurian, Permian, and Devonian geological time periods. Murchison’s findings were regarded as the crowning achievement of 19th-century geology and saw him elected president of the Geological Society in 1831.

Muybridge, Eadweard (1830–1904) English photographer known for his pioneering work in photographing motion. An established landscape photographer, Muybridge captured images of horses in full gallop by using up to 24 cameras and fast shutter speeds. He presented apparently moving images of animals using his zoopraxiscope.

Nakamura, Shuji, Japanese electronic engineer and inventor (1954–) See p.331

Napier, John (1550–1617) Scottish mathematician who invented logarithms. The 8th Laird of Merchistoun, Napier introduced the use of the decimal point in fractions, developed logarithms for mathematical calculations, and devised a set of calculating rods called Napier’s bones. He also designed secret weapons to defend Scotland against a perceived Catholic attack.

Newcomen, Thomas (1663–1729) English engineer and inventor of the first practical steam engine. Developed in conjunction with Thomas Savery, the Newcomen engine was originally used to pump water from a coal mine in Dudley, Staffordshire. Over the next 75 years, hundreds of Newcomen engines greatly increased coal production in England and contributed significantly to industrialization.

Newton, Isaac, English physicist and mathematician (1642–1727) See p.118

Nightingale, Florence (1820–1910) English nurse who reformed hospitals and founded modern nursing. Dubbed “the lady with the lamp” for her nursing work during the Crimean War, Nightingale founded the nurses training school at London’s St Thomas’ Hospital in 1861. She also helped improve public health in India and introduced new statistical techniques.

Nobel, Alfred (1833–96) Swedish chemist who invented dynamite, a less sensitive form of nitroglycerine, and founded the Nobel Prizes. He willed the majority of his fortune to creating the Nobel Prize, an annual award for achievements in physics, chemistry, medicine, literature, and peace.

Noether, Emmy (1882–1935) German mathematician and pioneering leader of abstract algebra. Appointed a lecturer at Göttingen University in 1919, she won acclaim